DIB isolators, EEPROM, ADC, MCU

File: DIB_EEPROM_ADC_MCU.kicad_sch

DAC, OE, Vref, IOEXP, Temp

File: DAC_Vref_IOEXP_Temp.kicad_sch

Power pre-regulator

File: Power_pre-regulator.kicad_sch

Bias power

File: Bias_power.kicad_sch

Post-regulator

File: Post-regulator.kicad_sch

Rsense protection, power output, Rprog

File: Rsense_protection_power_output.kicad_sch

Mode detection, OVP

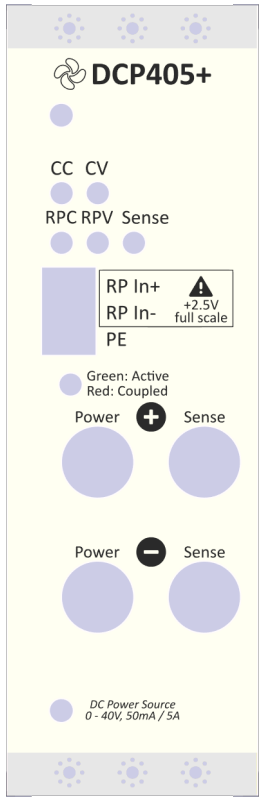
File: CC_CV_OVP.kicad_sch

Front mask details:

Material: Aluminum PCB
Thickness: 1.6 mm
PCB layout: 1-layer

DCP405+ project repository: https://github.com/eez-open/dib-dcp405plus/blob/main/KiCad/EEZ%20DIB%20DCP405plus%20front%20mask.kicad_pro
DCP405B+ project repository: https://github.com/eez-open/dib-dcp405plus/blob/main/KiCad/EEZ%20DIB%20DCP405plus%20front%20mask%20(Basic).kicad_pro





KK1 PCB-MB-01

KK2 PCB-MB-01

KK3 M3X8/D7985B

KK4 M3X8/D7985B

KK5 M3X8/D7985B

KK6 M3X8/D7985B

KK10 B3/BN2842

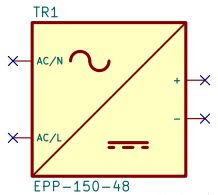
KK7 M3X8/D7985B

KK8 SK96/84SA

KK9 SP400-0.007-00-104

M1 TBP03P2-350-03BEOR

TR1



EPP-150-48

Alternative: Mean Well LOP-200-48

KK11 M3X4/D7985

KK12 M3X4/D7985

KK13 M3X4/D7985

KK14 M3X4/D7985

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Power module with SMPS pre-regulator, remote sense & programming

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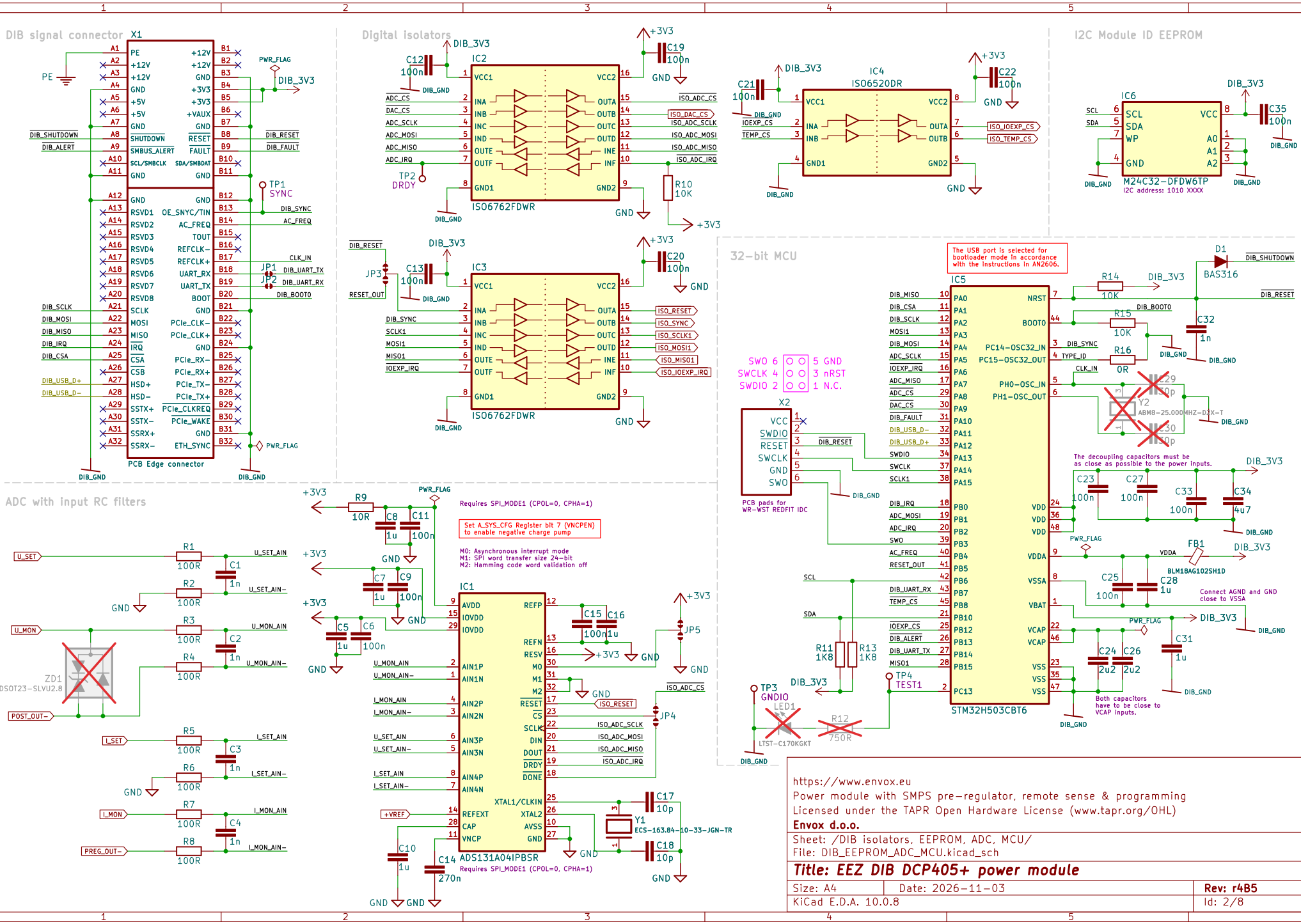
Envox d.o.o.

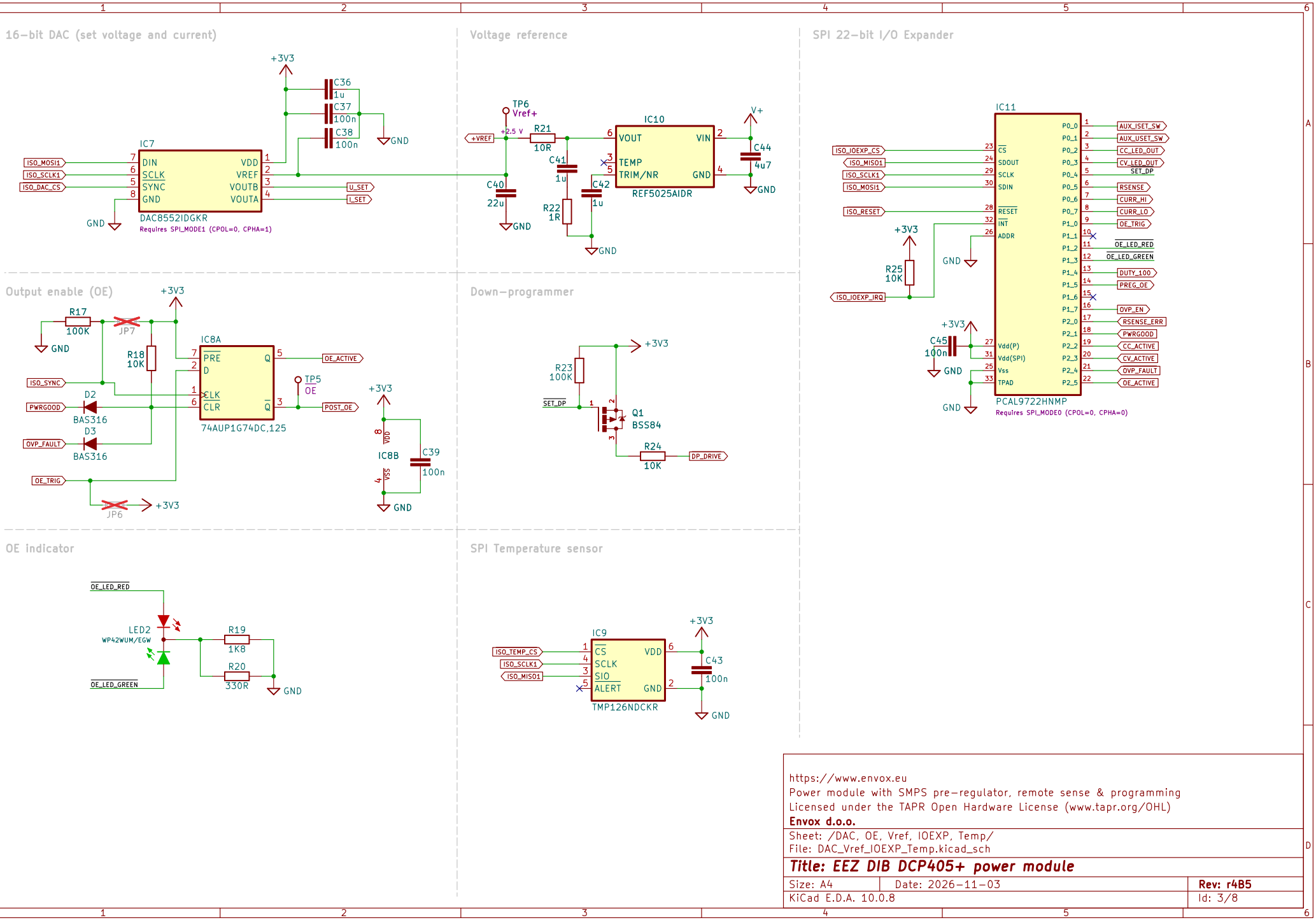
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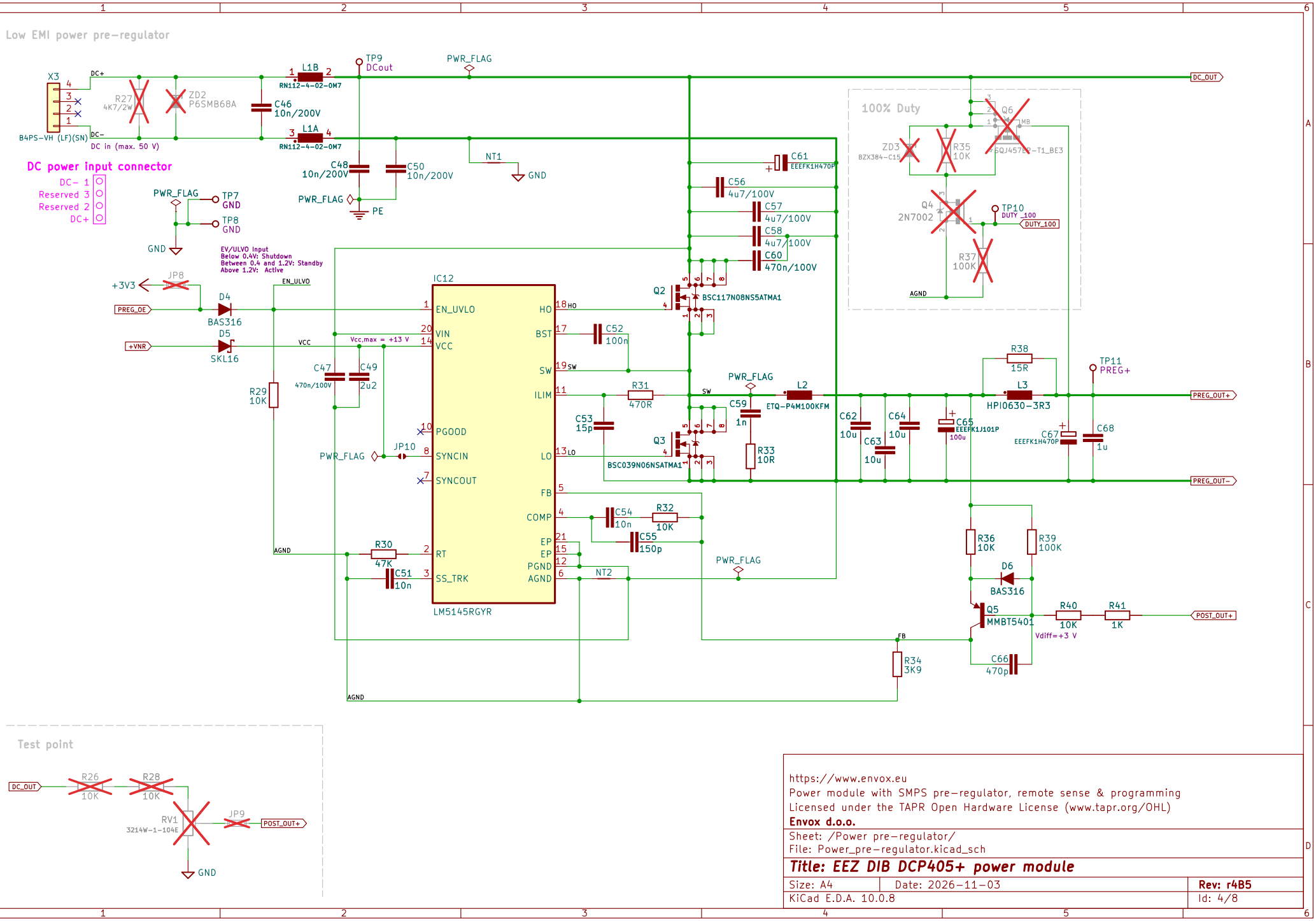
File: EEZ DIB DCP405plus.kicad_sch

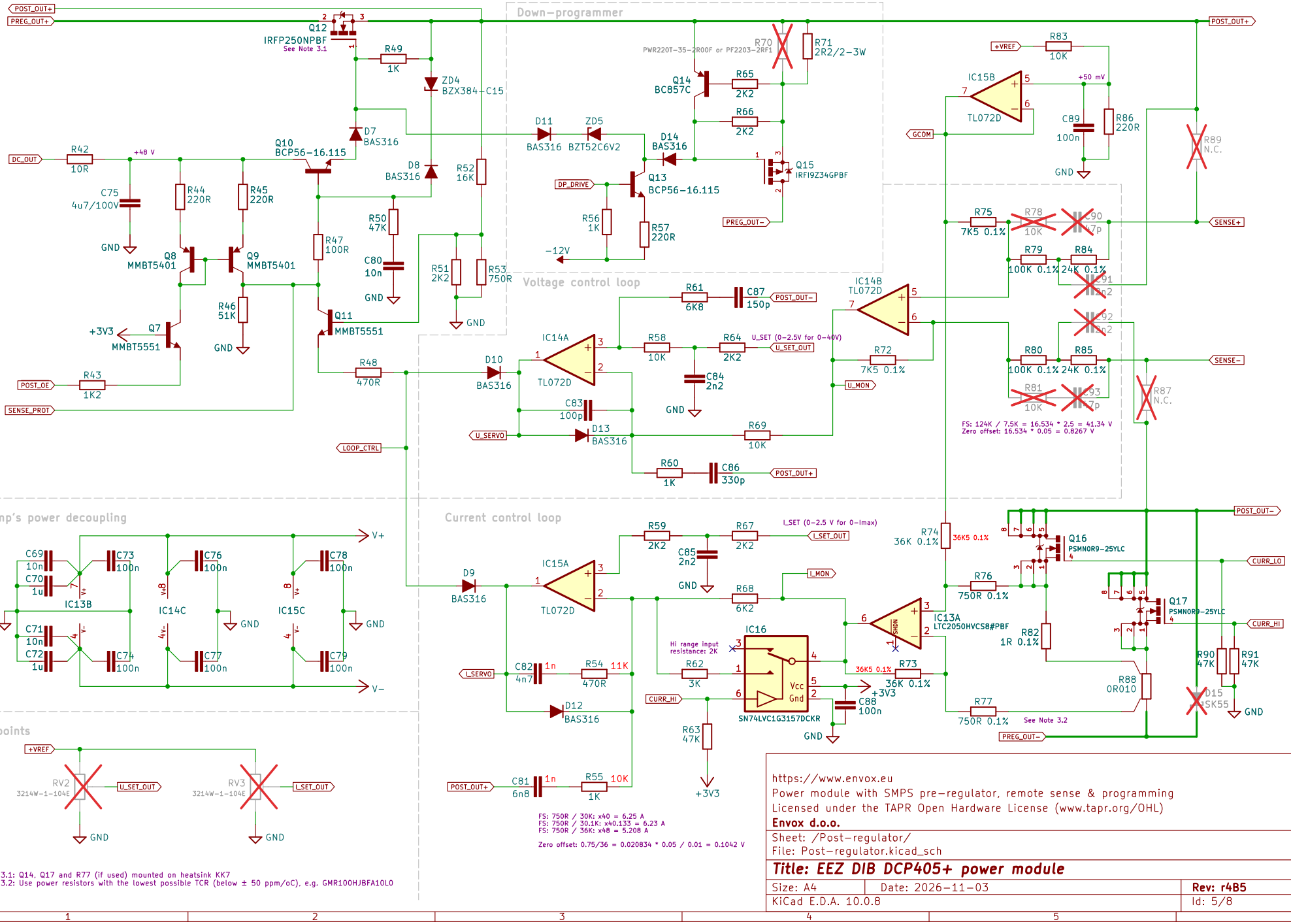
Title: EEZ DIB DCP405+ power module

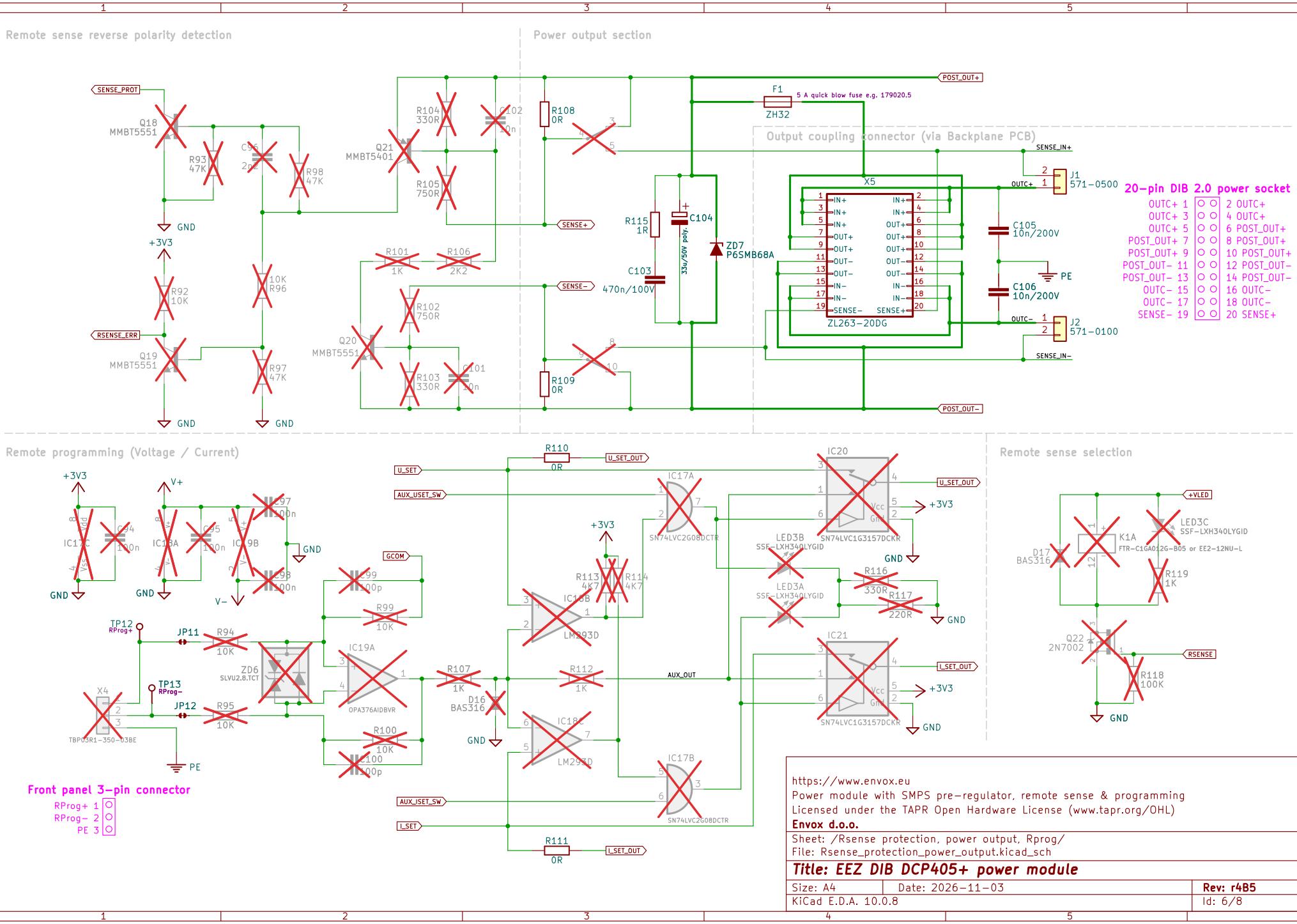
Size: A4	Date: 2026-11-03	**Rev: r4B5**
KiCad E.D.A. 10.0.8		Id: 1/8


DIB signal connector
X1
PCB Edge connector
Digital isolators
IC2
ISO6762FDWR
IC3
ISO6762FDWR
IC4
ISO6520DR
I2C Module ID EEPROM
IC6
M24C32-DFDW6TP
I2C address: 1010 XXXX
32-bit MCU
IC5
STM32H503CBT6
The USB port is selected for bootloader mode in accordance with the instructions in AN2606.
PCB pads for WR-WST REDFIT IDC
The decoupling capacitors must be as close as possible to the power inputs.
Connect AGND and GND close to VSSA
Both capacitors have to be close to VCAP inputs.
ADC with input RC filters
IC1
ADS131A04IPBSR
Requires SPI_MODE1 (CPOL=0, CPHA=1)
Set A_SYS_CFG Register bit 7 (VNCPEN) to enable negative charge pump
M0: Asynchronous interrupt mode
M1: SPI word transfer size 24-bit
M2: Hamming code word validation off
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Sheet: /DIB isolators, EEPROM, ADC, MCU/
File: DIB_EEPROM_ADC_MCU.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-11-03
Rev: r4B5
KiCad E.D.A. 10.0.8
Id: 2/8

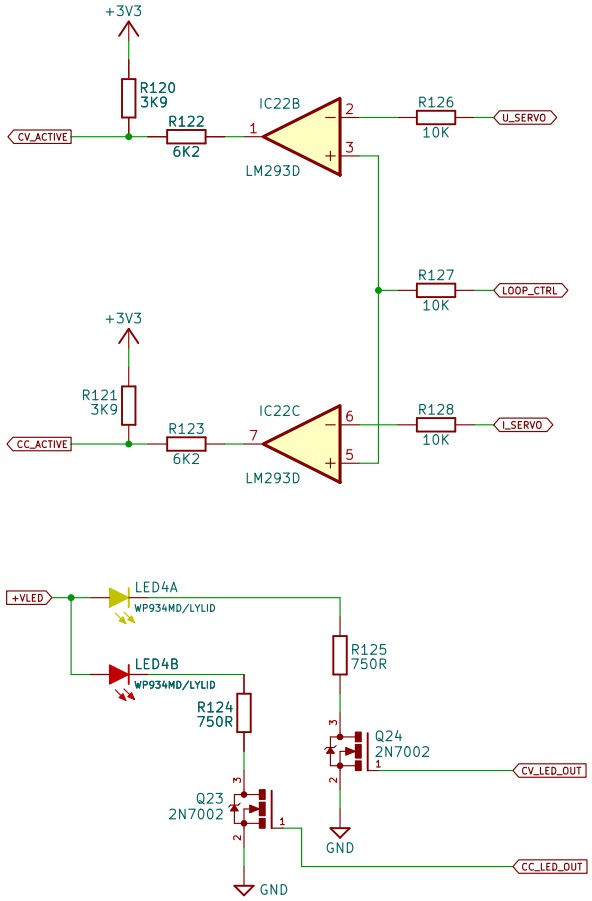

16-bit DAC (set voltage and current)
IC7
DAC8552IDGKR
Requires SPI_MODE1 (CPOL=0, CPHA=1)
Voltage reference
IC10
REF5025AIDR
SPI 22-bit I/O Expander
IC11
PCAL9722HNMP
Requires SPI_MODE0 (CPOL=0, CPHA=0)
Output enable (OE)
IC8A
IC8B
74AUP1G74DC,125
Down-programmer
OE indicator
SPI Temperature sensor
IC9
TMP126NDCKR
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Sheet: /DAC, OE, Vref, IOEXP, Temp/
File: DAC_Vref_IOEXP_Temp.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-11-03
Rev: r4B5
KiCad E.D.A. 10.0.8
Id: 3/8


Low EMI power pre-regulator
DC power input connector
B4PS-VH (LF)(SN)
DC in (max. 50 V)
DC- 1
Reserved 3
Reserved 2
DC+
X3
R27 4K7/2W
ZD2 P6SMB68A
C46 10n/200V
L1B RN112-4-02-0M7
L1A RN112-4-02-0M7
TP9 DCout
PWR_FLAG
C48 10n/200V
C50 10n/200V
PE
NT1
GND
TP7 GND
TP8 GND
JP8
+3V3
PREG_OE
D4 BAS316
+VNR
D5 SKL16
EN_UVLO
VCC
Vcc,max = +13 V
EV/ULVO Input
Below 0.4V: Shutdown
Between 0.4 and 1.2V: Standby
Above 1.2V: Active
C47 470n/100V
C49 2u2
R29 10K
IC12
EN_UVLO
VIN
VCC
PGOOD
SYNCIN
SYNCOUT
RT
SS_TRK
HO
BST
SW
ILIM
LO
FB
COMP
EP
EP
PGND
AGND
LM5145RGYR
JP10
R30 47K
C51 10n
C52 100n
R31 470R
C53 15p
C54 10n
R32 10K
C55 150p
NT2
AGND
Q2 BSC117N08NS5ATMA1
Q3 BSC039N06NSATMA1
C61 EEEFK1H470P
C56 4u7/100V
C57 4u7/100V
C58 4u7/100V
C60 470n/100V
C59 1n
R33 10R
L2 ETQ-P4M100KFM
C62 10u
C63 10u
C64 10u
C65 EEEFK1J101P 100u
C67 EEEFK1H470P
C68 1u
R38 15R
L3 HPI0630-3R3
TP11 PREG+
DC_OUT
PREG_OUT+
PREG_OUT-
100% Duty
ZD3 BZX384-C15
R35 10K
Q6 SQJ457EP-T1_BE3
Q4 2N7002
TP10 DUTY_100
R37 100K
R36 10K
R39 100K
D6 BAS316
Q5 MMBT5401
R40 10K
R41 1K
POST_OUT+
Vdiff=+3 V
R34 3K9
C66 470p
FB
Test point
R26 10K
R28 10K
RV1 3214W-1-104E
JP9
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Sheet: /Power pre-regulator/
File: Power_pre-regulator.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-11-03
Rev: r4B5
KiCad E.D.A. 10.0.8
Id: 4/8

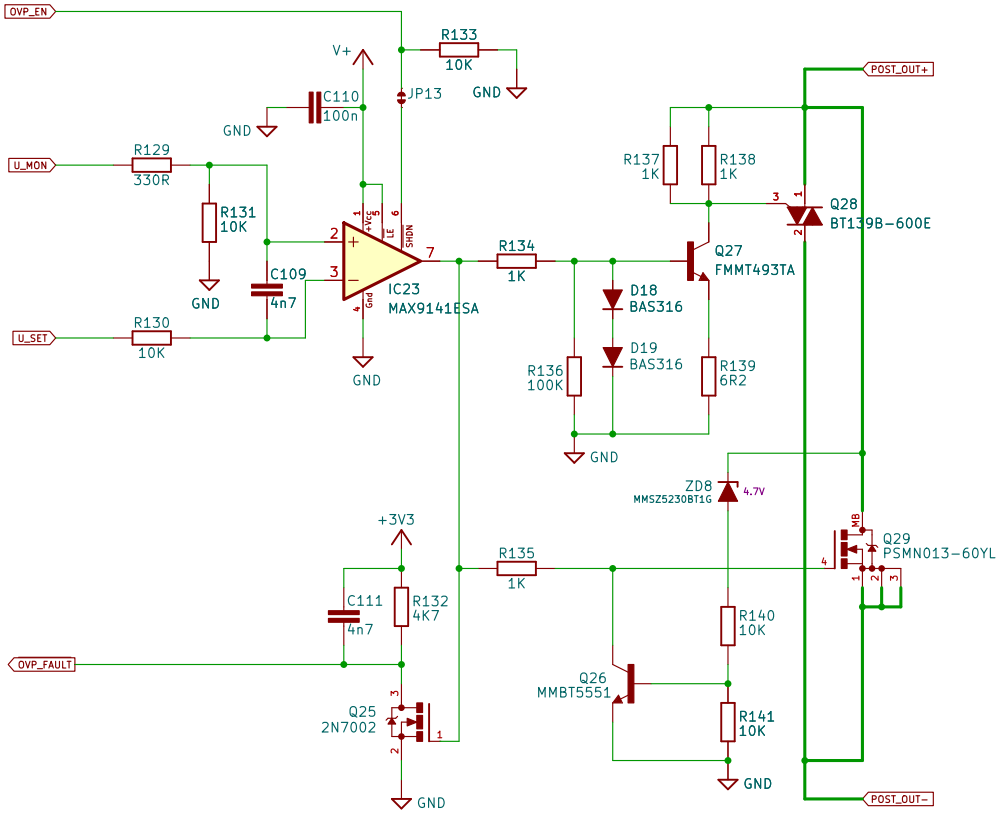

Down-programmer
Voltage control loop
Current control loop
Op amp's power decoupling
Test points
POST_OUT+
PREG_OUT+
DC_OUT
+48 V
Q12
IRFP250NPBF
See Note 3.1
Q10
BCP56-16.115
Q11
MMBT5551
Q7
MMBT5551
Q8
MMBT5401
Q9
MMBT5401
Q13
BCP56-16.115
Q14
BC857C
Q15
IRFI9Z34GPBF
Q16
PSMN0R9-25YLC
Q17
PSMN0R9-25YLC
D7
BAS316
D8
BAS316
D10
BAS316
D11
BAS316
D12
BAS316
D13
BAS316
D14
BAS316
D9
BAS316
ZD4
BZX384-C15
ZD5
BZT52C6V2
R70
PWR220T-35-2R00F or PF2203-2RF1
R71
2R2/2-3W
R42
10R
R43
1K2
R44
220R
R45
220R
R46
51K
R47
100R
R48
470R
R49
1K
R50
47K
R51
2K2
R52
16K
R53
750R
R54
470R
11K
R55
10K
1K
R56
1K
R57
220R
R58
10K
R59
2K2
R60
1K
R61
6K8
R62
3K
R63
47K
R64
2K2
R65
2K2
R66
2K2
R67
2K2
R68
6K2
R69
10K
R72
7K5 0.1%
R73
36K 0.1%
R74
36K 0.1%
36K5 0.1%
R75
7K5 0.1%
R76
750R 0.1%
R77
750R 0.1%
See Note 3.2
R78
10K
R79
100K 0.1%
R80
100K 0.1%
R81
10K
R82
1R 0.1%
R83
10K
R84
24K 0.1%
R85
24K 0.1%
R86
220R
R87
N.C.
R88
0R010
R89
N.C.
R90
47K
R91
47K
C69
10n
C70
1u
C71
10n
C72
1u
C73
100n
C74
100n
C75
4u7/100V
C76
100n
C77
100n
C78
100n
C79
100n
C80
10n
C81
6n8
1n
C82
4n7
1n
C83
100p
C84
2n2
C85
2n2
C86
330p
C87
150p
C88
100n
C89
100n
C90
47p
C91
2n2
C92
2n2
C93
47p
IC13B
IC14C
IC15C
IC14A
TL072D
IC15B
TL072D
IC14B
TL072D
IC13A
LTC2050HVCS8#PBF
IC16
SN74LVC1G3157DCKR
RV2
3214W-1-104E
RV3
3214W-1-104E
D15
1SK55
U_SET (0-2.5V for 0-40V)
I_SET (0-2.5 V for 0-Imax)
Hi range input resistance: 2K
FS: 124K / 7.5K = 16.534 * 2.5 = 41.34 V
Zero offset: 16.534 * 0.05 = 0.8267 V
FS: 750R / 30K: x40 = 6.25 A
FS: 750R / 30.1K: x40.133 = 6.23 A
FS: 750R / 36K: x48 = 5.208 A
Zero offset: 0.75/36 = 0.020834 * 0.05 / 0.01 = 0.1042 V
Note 3.1: Q14, Q17 and R77 (if used) mounted on heatsink KK7
Note 3.2: Use power resistors with the lowest possible TCR (below ± 50 ppm/oC), e.g. GMR100HJBFA10L0
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Sheet: /Post-regulator/
File: Post-regulator.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-11-03
Rev: r4B5
KiCad E.D.A. 10.0.8
Id: 5/8


Remote sense reverse polarity detection
Power output section
Output coupling connector (via Backplane PCB)
Remote programming (Voltage / Current)
Remote sense selection
SENSE_PROT
RSENSE_ERR
Q18 MMBT5551
Q19 MMBT5551
Q20 MMBT5551
Q21 MMBT5401
R92 10K
R93 47K
R96 10K
R97 47K
R98 47K
R101 1K
R102 750R
R103 330R
R104 330R
R105 750R
R106 2K2
C96 2n2
C101 10n
C102 10n
R108 0R
R109 0R
SENSE+
SENSE-
R115 1R
C103 470n/100V
C104 33u/50V poly.
ZD7 P6SMB68A
F1
ZH32
5 A quick blow fuse e.g. 179020.5
POST_OUT+
POST_OUT-
SENSE_IN+
SENSE_IN-
X5
ZL263-20DG
J1 571-0500
J2 571-0100
OUTC+
OUTC-
C105 10n/200V
C106 10n/200V
PE
20-pin DIB 2.0 power socket
OUTC+ 1
OUTC+ 3
OUTC+ 5
POST_OUT+ 7
POST_OUT+ 9
POST_OUT- 11
POST_OUT- 13
OUTC- 15
OUTC- 17
SENSE- 19
2 OUTC+
4 OUTC+
6 POST_OUT+
8 POST_OUT+
10 POST_OUT+
12 POST_OUT-
14 POST_OUT-
16 OUTC-
18 OUTC-
20 SENSE+
+3V3
V+
V-
GND
IC17C
IC18A
IC19B
C94 100n
C95 100n
C97 100n
C98 100n
GCOM
C99 100p
R99 10K
IC19A
OPA376AIDBVR
R100 10K
C100 100p
TP12 RProg+
TP13 RProg-
JP11
JP12
R94 10K
R95 10K
ZD6 SLVU2.8.TCT
X4
TBP03R1-350-03BE
Front panel 3-pin connector
RProg+ 1
RProg- 2
PE 3
U_SET
AUX_USET_SW
I_SET
AUX_ISET_SW
R110 0R
R111 0R
U_SET_OUT
I_SET_OUT
IC17A
SN74LVC2G08DCTR
IC17B
IC18B
LM293D
R113 4K7
R114 4K7
R107 1K
R112 1K
D16 BAS316
AUX_OUT
IC20
SN74LVC1G3157DCKR
IC21
LED3B SSF-LXH340LYGID
LED3A SSF-LXH340LYGID
R116 330R
R117 220R
Remote sense selection
+VLED
K1A FTR-C1GA012G-B05 or EE2-12NU-L
LED3C SSF-LXH340LYGID
D17 BAS316
R119 1K
Q22 2N7002
RSENSE
R118 100K
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Sheet: /Rsense protection, power output, Rprog/
File: Rsense_protection_power_output.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-11-03
Rev: r4B5
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Id: 6/8

Constant Voltage (CV) and Constant Current (CC) mode detection and indication



Over-voltage protection (OVP)



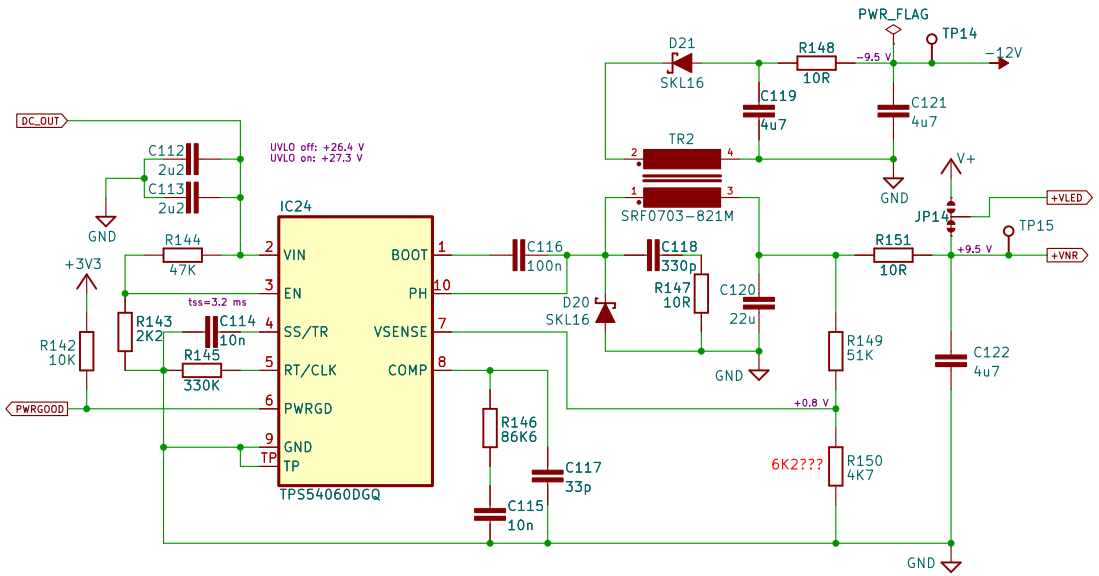
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Sheet: /Mode detection, OVP/
File: CC_CV_OVP.kicad_sch

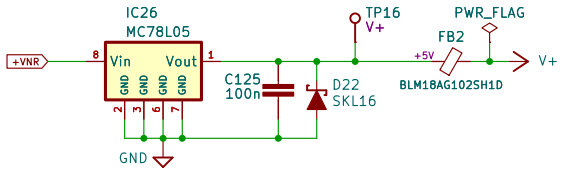
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Size: A4	Date: 2026-11-03	**Rev: r4B5**
KiCad E.D.A. 10.0.8		Id: 7/8

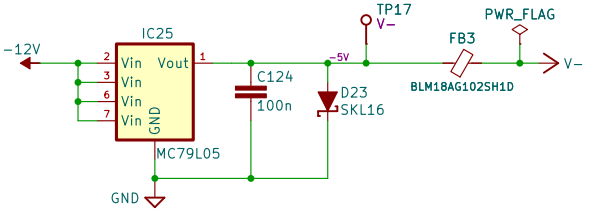
Bias power supply



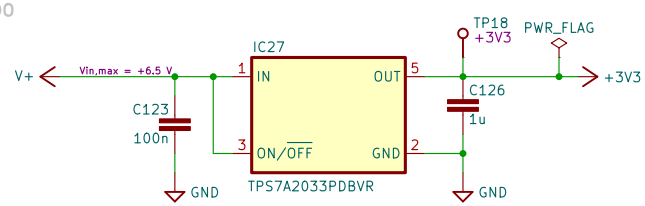
+5V LDO



-5V LDO



+3V3 LDO



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Sheet: /Bias power/
File: Bias_power.kicad_sch

Title: EEZ DIB DCP405+ power module

Size: A4	Date: 2026-11-03	**Rev: r4B5**
KiCad E.D.A. 10.0.8		Id: 8/8